Chapter 1 – Introduction

1.1 Basic idea

- To design the *best possible* structure.
- *Design variables*.
- *Objective function*.
- *Constraints*.

1.2 The (structural) design process

– Function (design brief).
– Conceptual design.
– Optimization: iterative-intuitive vs. mathematical design optimization approach.
– Details.

1.3 General mathematical form of a struct. opt. problem

– Simultaneous analysis and design (SAND) approach:

$$\begin{aligned} \min_{\mathbf{x},\mathbf{u}} : \quad & f(\mathbf{x},\mathbf{u}) \\ \text{s.t.} : \quad & g_i(\mathbf{x},\mathbf{u}) \leq 0, \quad i = 1,\ldots,l \\ & \mathbf{K}(\mathbf{x})\mathbf{u} = \mathbf{F}(\mathbf{x}) \end{aligned}$$

– Nested approach (based on elimination of $\mathbf{u}$):

$$\begin{aligned} \min_{\mathbf{x}} : \quad & f(\mathbf{x},\mathbf{u}(\mathbf{x})) \\ \text{s.t.} : \quad & g_i(\mathbf{x},\mathbf{u}(\mathbf{x})) \leq 0, \quad i = 1,\ldots,l \end{aligned}$$

where $\mathbf{u}(\mathbf{x}) = \mathbf{K}^{-1}(\mathbf{x})\mathbf{F}(\mathbf{x})$

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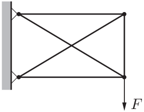
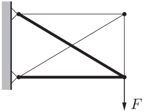
1.4 Three types of structural optimization problems

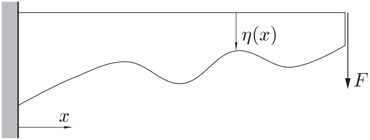
- Size optimization
- Shape optimization
- Topology optimization (truss topology design vs. continuum topology optimization)

Initial design

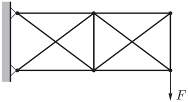


Optimized design

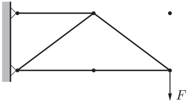


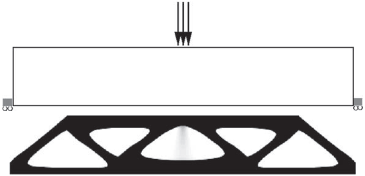


Initial design



Optimized design





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1.5 Discrete and distributed parameter systems

– Discrete: $\mathbf{x}$ and $\mathbf{u}$ are vectors

– Distributed: x and u are spatial fields: must be discretized
Convergence, existence of solutions: difficult to prove → engineering judgment

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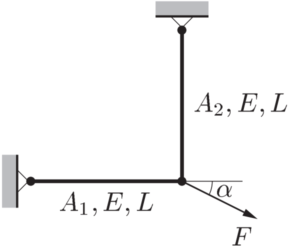
Chapter 2 – Examples of optimization of discrete parameter systems

2.1 Two-bar truss s.t. stress constraints

Minimize the weight of a two-bar truss subject to stress constraints. The design variables are the bar areas A_1 and A_2, the material density is ρ, the maximum allowable stress is σ_0.



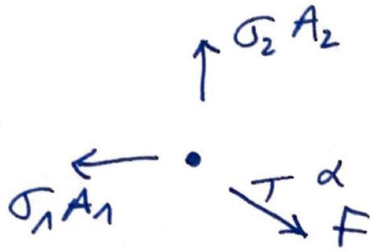
Parameter values for numerical solution:

$$
\begin{aligned}
\alpha &= 30^\circ \\
L &= 1\,\mathrm{m} \\
\rho &= 7850\,\mathrm{kg/m^3} \\
\sigma_0 &= 235\,\mathrm{MPa} \\
F &= 200\,\mathrm{kN}
\end{aligned}
$$

- Solve the problem analytically (`ex1a.m`).
- Solve the problem using the method of moving asymptotes or MMA (`ex1b.m`).
- Solve the problem using particle swarm optimization or PSO (`ex1c.m`).

2.1 Two-bar truss subject to stress constraints

As the structure is statically determinate, the stresses can be written immediately in terms of the design variables A_1 and A_2:



$$\sigma_1 = \frac{F}{A_1} \cos\alpha$$

$$\sigma_2 = \frac{F}{A_2} \sin\alpha$$

Formulation of the optimization problem:

$$\min_{A_1, A_2} \quad \rho L (A_1 + A_2)$$

$$\text{s.t.} \quad \frac{F}{A_1} \cos\alpha \leq \sigma_0$$

$$\frac{F}{A_2} \sin\alpha \leq \sigma_0$$

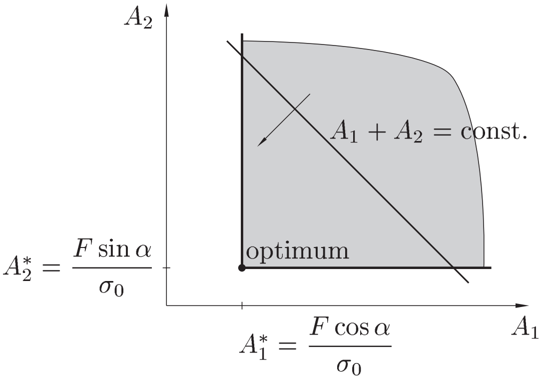
Or:

$$\min_{A_1, A_2} \quad A_1 + A_2$$

$$\text{s.t.} \quad A_1 \geq \frac{F}{\sigma_0} \cos\alpha$$

$$A_2 \geq \frac{F}{\sigma_0} \sin\alpha$$

2.1 Two-bar truss s.t. stress constraints



Solution with MMA / PSO:

$$\min_{x} f_0(x)$$
$$\text{s.t. } f_i(x) \leq 0$$

$$f_0(x) = \rho L (x_1 + x_2)$$
$$f_1(x) = \frac{F \cos \alpha}{x_1 \sigma_0} - 1$$
$$f_2(x) = \frac{F \sin \alpha}{x_2 \sigma_0} - 1$$

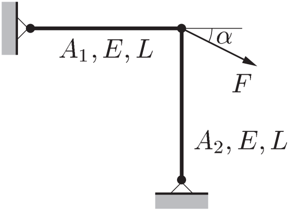
$$\frac{df_0}{dx_1} = \rho L \qquad \frac{df_0}{dx_2} = \rho L$$

$$\frac{df_1}{dx_1} = -\frac{F \cos \alpha}{x_1^2 \sigma_0} \qquad \frac{df_1}{dx_2} = 0$$

$$\frac{df_2}{dx_1} = 0 \qquad \frac{df_2}{dx_2} = -\frac{F \sin \alpha}{x_2^2 \sigma_0}$$

2.2 Two-bar truss s.t. stress and instability constraints

Minimize the weight of a two-bar truss subject to stress constraints and instability constraints. The design variables are the bar areas A_1 and A_2, the material density is ρ, the maximum allowable stress is σ_0, the Young's modulus is E, the bars have circular cross sections, and the buckling load is 1/4 of the theoretical Euler buckling load. The angle α is fixed at $45°$.



Parameter values for numerical solution:

$$
\begin{aligned}
L &= 1\,\mathrm{m} \\
\rho &= 7850\,\mathrm{kg/m^3} \\
\sigma_0 &= 235\,\mathrm{MPa} \\
E &= 210\,\mathrm{GPa} \\
F &= 200\,\mathrm{kN}
\end{aligned}
$$

- Solve the problem analytically (`ex2a.m`).
- Solve the problem using MMA (`ex2b.m`).
- Solve the problem using PSO (`ex2c.m`).

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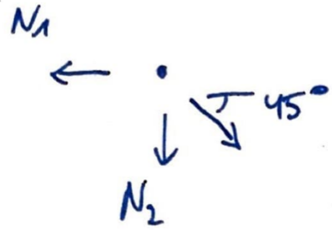
2.2 Two-bar truss subject to stress and instability constraints



$$\sigma_1 = \frac{F}{\sqrt{2}\, A_1}$$

$$\sigma_2 = \frac{-F}{\sqrt{2}\, A_2}$$

$$N_2 = \frac{-F}{\sqrt{2}}$$

Constraints :

$$\sigma_1 \leq \sigma_0$$

$$-\sigma_2 \leq \sigma_0$$

$$-N_2 \leq \frac{1}{4} P_c = \frac{1}{4} \pi^2 \frac{EI}{L^2}$$

↑ safety factor

where $I = \dfrac{A_2^2}{4\pi}$

$\left(I = \dfrac{\pi D^4}{64} \right)$

Solution with MMA/PSO :

$$\min_{x} \; f_0(x)$$
$$\text{s.t.} \; f_i(x) \leq 0$$

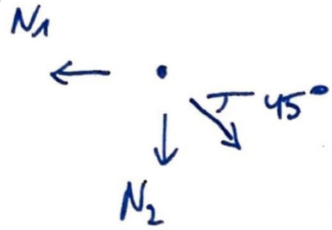
$f_0(x) =$

$f_1(x) =$

$f_2(x) =$

$f_3(x) =$

2.2 Two-bar truss subject to stress and instability constraints



$$\sigma_1 = \frac{F}{\sqrt{2}\,A_1}$$

$$\sigma_2 = \frac{-F}{\sqrt{2}\,A_2}$$

$$N_2 = \frac{-F}{\sqrt{2}}$$

Constraints :

$$\sigma_1 \leq \sigma_0$$

$$-\sigma_2 \leq \sigma_0$$

$$-N_2 \leq \frac{1}{4} P_c = \frac{1}{4} \pi^2 \frac{EI}{L^2}$$

safety factor

where $I = \dfrac{A_2^2}{4\pi}$

$\left(I = \dfrac{\pi D^4}{64} \right)$

Solution with MMA/PSO :

$$\min_{x} \; f_0(x)$$

$$\text{s.t.} \; f_i(x) \leq 0$$

$$f_0(x) = \rho L (x_1 + x_2)$$

$$f_1(x) = \frac{F}{\sqrt{2}\, x_1 \sigma_0} - 1$$

$$f_2(x) = \frac{F}{\sqrt{2}\, x_2 \sigma_0} - 1$$

$$f_3(x) = \frac{16 F L^2}{\sqrt{2}\, \pi E x_2^2} - 1$$

$$\frac{df_0}{dx_1} = \rho L \qquad \frac{df_0}{dx_2} = \rho L$$

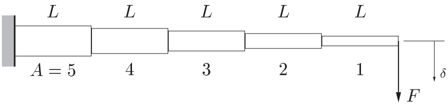
$$\frac{df_1}{dx_1} = -\frac{F}{\sqrt{2}\, x_1^2\, \sigma_0} \qquad \frac{df_1}{dx_2} = 0$$

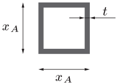
$$\frac{df_2}{dx_1} = 0 \qquad \frac{df_2}{dx_2} = -\frac{F}{\sqrt{2}\, x_2^2\, \sigma_0}$$

$$\frac{df_3}{dx_1} = 0 \qquad \frac{df_3}{dx_2} = -\frac{32\, F L^2}{\sqrt{2}\, \pi\, E\, x_2^3}$$

2.4 Cantilever beam s.t. displacement constraint

Minimize the weight of a cantilever beam consisting of N segments with square hollow sections subject to a displacement constraint. The design variables are the side lengths x_A of each segment A. The maximum allowable displacement is δ_0.





Cross-section of segment A.

Parameter values for numerical solution:

$$
\begin{aligned}
NL &= 1\,\mathrm{m} \quad \text{(total length is } 1\,\mathrm{m}\text{)} \\
t &= 1\,\mathrm{cm} \\
\rho &= 7850\,\mathrm{kg/m^3} \\
E &= 210\,\mathrm{GPa} \\
F &= 10\,\mathrm{kN} \\
\delta_0 &= 4\,\mathrm{mm}
\end{aligned}
$$

- Solve the problem analytically for $N = 2$ segments (`ex3a.m`).
- Solve the problem for $N = 2$ segments using MMA (`ex3b.m`).
- Solve the problem for $N = 2$ segments using PSO (`ex3c.m`).
- Solve the problem for $N = 100$ segments using MMA and PSO (`ex3b.m` and `ex3c.m`).

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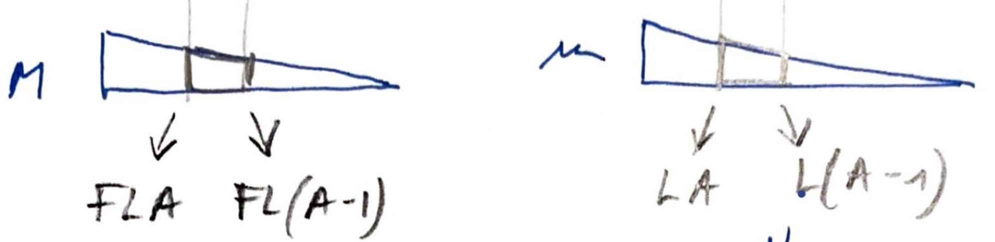
2.4 Cantilever beam s.t. displacement constraint

* displacement δ via virtual work :



$$\delta = \sum_{A=1}^{N} \frac{1}{EI_A} \int M m \, dx = \sum_{A=1}^{N} \frac{1}{EI_A} \left(\square \times \square \right)$$

$$= \sum_{A=1}^{N} \frac{1}{EI_A} \frac{1}{6} \left(2 i_1 k_1 + i_1 k_2 + i_2 k_1 + 2 i_2 k_2 \right) \ell$$

$$= \sum_{A=1}^{N} \frac{1}{EI_A} \frac{1}{6} \left(2FL^2A^2 + 2FL^2 A (A-1) + 2FL^2 (A-1)^2 \right) L$$

Newton's binomial:

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1

$$(a+b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$$

$$= \sum_{A=1}^{N} \frac{FL^3}{EI_A} \left(A^2 - A + \frac{1}{3} \right)$$

$$I_A = \frac{x_A^4}{12} - \frac{(x_A - 2t)^4}{12}$$

$$\approx \frac{x_A^4}{12} - \frac{x_A^4}{12} + \frac{4 x_A^3 \, 2t}{12} = \frac{2 t x_A^3}{3}$$

$$\Rightarrow \delta = \frac{3FL^3}{2Et} \sum_{A=1}^{N} \left(A^2 - A + \frac{1}{3} \right) \frac{1}{x_A^3}$$

* weight :

$$W = L\rho \sum_{A=1}^{N} \left(x_A^2 - (x_A - 2t)^2 \right) \approx 4 L \rho t \sum_{A=1}^{N} x_A$$

Formulation of the opt. problem for $N = 2$:

$$\min \quad 4L\rho t (x_1 + x_2)$$

$$\text{s.t.} \quad \frac{3FL^3}{2Et}\left(\frac{1}{3}\frac{1}{x_1^3} + \frac{7}{3}\frac{1}{x_2^3}\right) \leq \delta_0 .$$

Or:

$$\min \quad C_1 (x_1 + x_2)$$

$$\text{s.t.} \quad \frac{1}{x_1^3} + \frac{7}{x_2^3} \leq C_2$$

where $C_1 = 4L\rho t$

$$C_2 = \frac{2Et\delta_0}{3FL^3}$$

Analytical solution

Assume that the displacement constraint is active:

$$\frac{1}{x_1^3} + \frac{7}{x_2^3} = C_2$$

$$\Rightarrow \quad x_1 = \left(C_2 - 7x_2^{-3}\right)^{-\frac{1}{3}}$$

$$\Rightarrow \quad W = C_1\left[\left(C_2 - 7x_2^{-3}\right)^{-\frac{1}{3}} + x_2\right]$$

Find stationary point of obj fun:

$$\frac{dW}{dx_2} = C_1\left[-7\left(C_2 x_2^3 - 7\right)^{-\frac{4}{3}} + 1\right] = 0$$

$$\Rightarrow \quad x_2 = 7^{1/4}\left(\frac{1 + 7^{1/4}}{C_2}\right)^{\frac{1}{3}}$$

$$\Rightarrow \quad x_1 = \left(\frac{1 + 7^{1/4}}{C_2}\right)^{\frac{1}{3}}$$

Solution (for N segments) with MMA/PSO

$$\min \; f_0(x)$$
$$\text{s.t.} \; f_i(x) \leq 0$$

$$f_0(x) = 4\rho L t \sum_{A=1}^{N} x_A$$

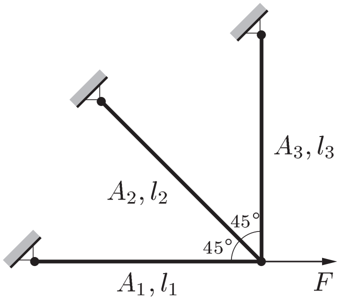
$$f_1(x) = \frac{\frac{3FL^3}{2Et} \sum_{A=1}^{N} \left(A^2 - A + \frac{1}{3}\right) \frac{1}{x_A{}^3}}{\delta_0} - 1$$

$$\frac{df_0}{dx_A} = 4\rho L t$$

$$\frac{df_1}{dx_A} = \frac{-\frac{9FL^3}{2Et} \left(A^2 - A + \frac{1}{3}\right) \frac{1}{x_A{}^4}}{\delta_0}$$

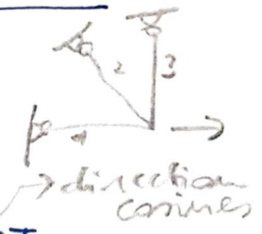
2.5 Three-bar truss s.t. stress constraints

Minimize the weight of a three-bar truss subject to stress constraints. Bars 1 and 3 are assumed to have the same section area $A_1 = A_3$, such that the number of design variables is reduced to 2: A_1 and A_2. The material density (or the cost per unit volume) of bar i is ρ_i and the maximum allowable stress for bar i is σ_i. All bars have the same length $l_1 = l_2 = l_3 = L$.



– Solve the problem graphically for various combinations of the parameters ρ_i and σ_i (see textbook for the parameter values).

2.5 Three-bar truss subject to stress constraints



The structure is no longer statically determinate, so we need

- equilibrium equations $\quad F = B^T s$
- stress-strain relations $\quad s = D\,\delta$
- strain-displacement relations $\quad \delta = B u$

$\Rightarrow \quad F = B^T D B u \equiv K u \quad$: finite element approach

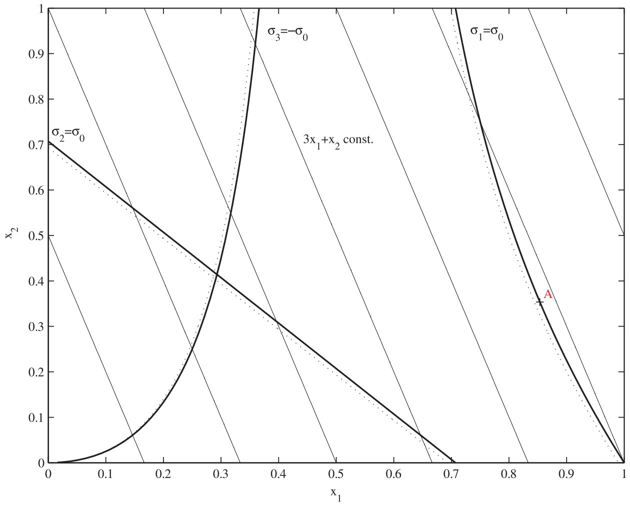
$\rightarrow$ determine analytical expressions for $\sigma_1, \sigma_2, \sigma_3$ in terms of A_1, A_2, $A_3 = A_1$.

2.5 Three-bar truss s.t. stress constraints

$$
\begin{aligned}
\min_{A_1, A_2} : \quad & (\rho_1 A_1 + \rho_2 A_2 + \rho_3 A_1) L \\
\text{s.t.} : \quad & \frac{F(2A_1 + A_2)}{2A_1(A_1 + A_2)} - \sigma_1^{\max} \leq 0 \\
& \frac{F}{\sqrt{2}(A_1 + A_2)} - \sigma_2^{\max} \leq 0 \qquad \text{if } A_2 > 0 \\
& \frac{F A_2}{2A_1(A_1 + A_2)} - \sigma_3^{\max} \leq 0 \\
& A_1 > 0 \\
& A_2 \geq 0
\end{aligned}
$$

2.5 Three-bar truss s.t. stress constraints

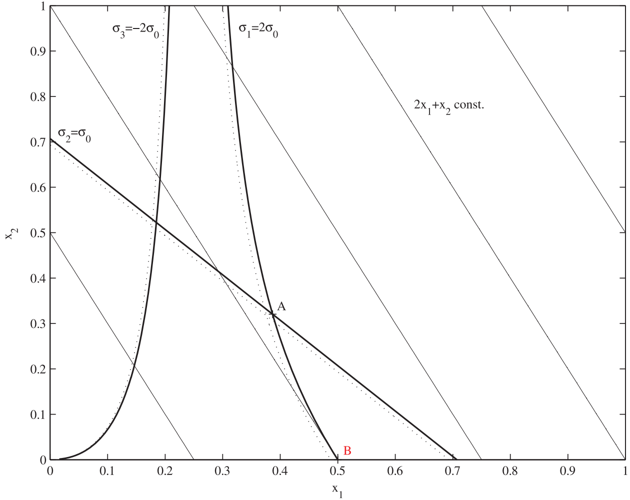
Case A: $\rho_1 = 2\rho_0, \quad \rho_2 = \rho_3 = \rho_0, \quad \sigma_1^{\max} = \sigma_2^{\max} = \sigma_3^{\max} = \sigma_0$



$$x_1 = \frac{A_1\sigma_0}{F}, \quad x_2 = \frac{A_2\sigma_0}{F}$$

2.5 Three-bar truss s.t. stress constraints

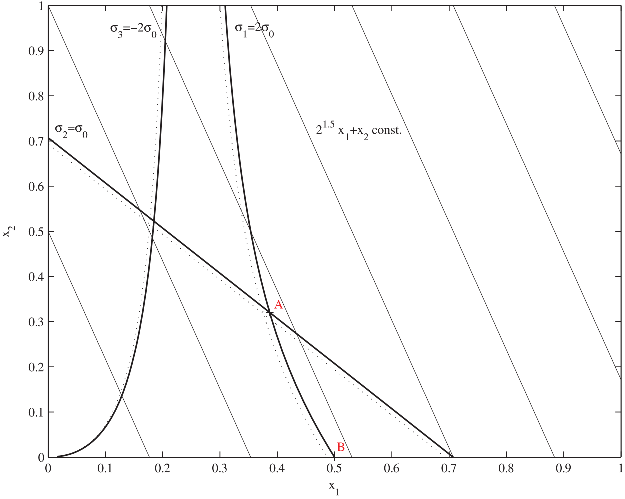
Case B: $\rho_1 = \rho_2 = \rho_3 = \rho_0, \quad \sigma_1^{\max} = \sigma_3^{\max} = 2\sigma_0, \quad \sigma_2^{\max} = \sigma_0$



$$x_1 = \frac{A_1\sigma_0}{F}, \quad x_2 = \frac{A_2\sigma_0}{F}$$

2.5 Three-bar truss s.t. stress constraints

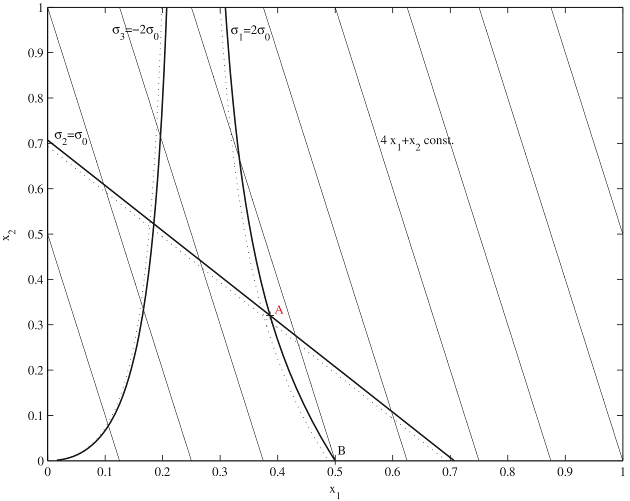
Case C: $\rho_1 = (2\sqrt{2}-1)\rho_0,\ \rho_2 = \rho_3 = \rho_0,\ \sigma_1^{\max} = \sigma_3^{\max} = 2\sigma_0,\ \sigma_2^{\max} = \sigma_0$



$$x_1 = \frac{A_1\sigma_0}{F}, \quad x_2 = \frac{A_2\sigma_0}{F}$$

2.5 Three-bar truss s.t. stress constraints

Case D: $\rho_1 = 3\rho_0, \quad \rho_2 = \rho_3 = \rho_0, \quad \sigma_1^{\max} = \sigma_3^{\max} = 2\sigma_0, \quad \sigma_2^{\max} = \sigma_0$



$$x_1 = \frac{A_1\sigma_0}{F}, \quad x_2 = \frac{A_2\sigma_0}{F}$$

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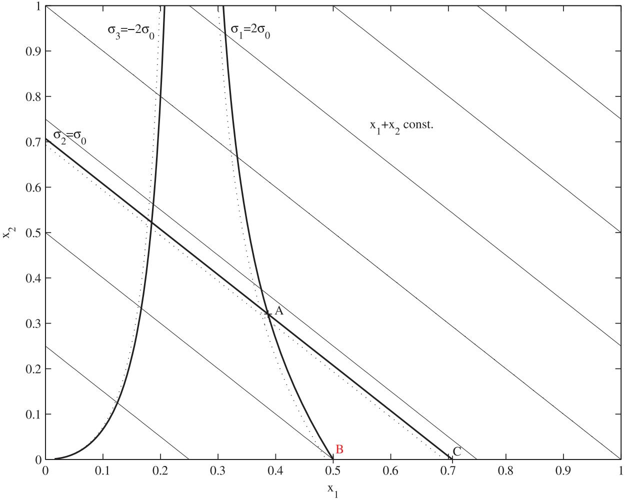
2.5 Three-bar truss s.t. stress constraints

Case E: $\rho_1 = \rho_3 = \rho_0, \quad \rho_2 = 2\rho_0, \quad \sigma_1^{\max} = \sigma_3^{\max} = 2\sigma_0, \quad \sigma_2^{\max} = \sigma_0$



$$x_1 = \frac{A_1\sigma_0}{F}, \quad x_2 = \frac{A_2\sigma_0}{F}$$